MSc thesis Proposal 2026-2027

Pedagogically relevant assessment of RAG-enhanced AI educational platform

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Objectives: RAGAS is a popular approach to assess a course-grounded tutoring platform. However, the automated RAGAS evaluation scores do not necessarily establish that an answer is pedagogically appropriate or helpful for learning. The goal of this thesis will be to develop a methodology to evaluate the pedagogical quality of the RAG-enhanced LLM tutoring responses. The methodology will be integrated and tested in the ChatToCaseStudy platform.

Research question: How reliably do RAGAS faithfulness and answer-relevance metrics assess the quality of course-grounded LLM tutoring responses, particularly when an answer is factually correct but pedagogically inadequate. The central hypothesis is that a response can receive high RAGAS scores while failing to help a student understand a concept, address a misconception, or reason independently.This evaluation study requires carefully designed examples, expert annotation, and quantitative validation.

Work plan

Task 1 Literature analysis
State of the art review of existing AI educational platform. Do they use methods and metrics to evaluate the pedagogical effectiveness ?

Task 2 Data (Q&A) generation.
Formulate about 100 course-related questions. Use an LLM to generate responses under different conditions, e.g. Good tutoring response, Grounded but pedagogically weak, Flawed response.

Task 3 Human (course instructor) in the loop
Create functionality on the ChatToCaseStudy platform for the instructor to access the questions of the students and the obtained answers, to correct the answers (if necessary) and store the checked Q&A samples in the so called Reference (golden) context database for this course.

Task 4 Test the ChatToCaseStudy platform with randomly selected Q&A.
Compute the RAGAS metrics, make statistical conclusions.

Task 5 Dissertation and Documentation.
Final stage involves writing the master dissertation and additional documentation, such as code repositories and reports, to ensure reproducibility and facilitate future research. Publishing the outcomes.

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